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Data Mining to Understand the Impact of the "Circadian Advantage" in the NHL

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Data mining to understand the impact of the "circadian advantage" in the NHL

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Course Code: DSE4800
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<u>Coefficient</u>	<u>Estimate</u>	<u>P-Value</u>
<u>Distance (Miles)</u>	-2.480e-04	0.00837
<u>Time Zone Game to Game Difference</u>	1.426e-01	0.05266
<u>Days Gap</u>	6.251e-02	0.05525

Introduction

I am data mining and feature engineering for two datasets. The datasets are from the 2016-17 and 2017-18 NHL Seasons. I created multiple graphs and ran several different tests. Those tests include T-tests, Correlation and Logistic Regression.

Objective

- To understand how travel between time zones affects athletes.
- If traveling from East to West affects athletes different than traveling West to East.
- Use the data collected to optimize team travel and improve player performance.

Methodology

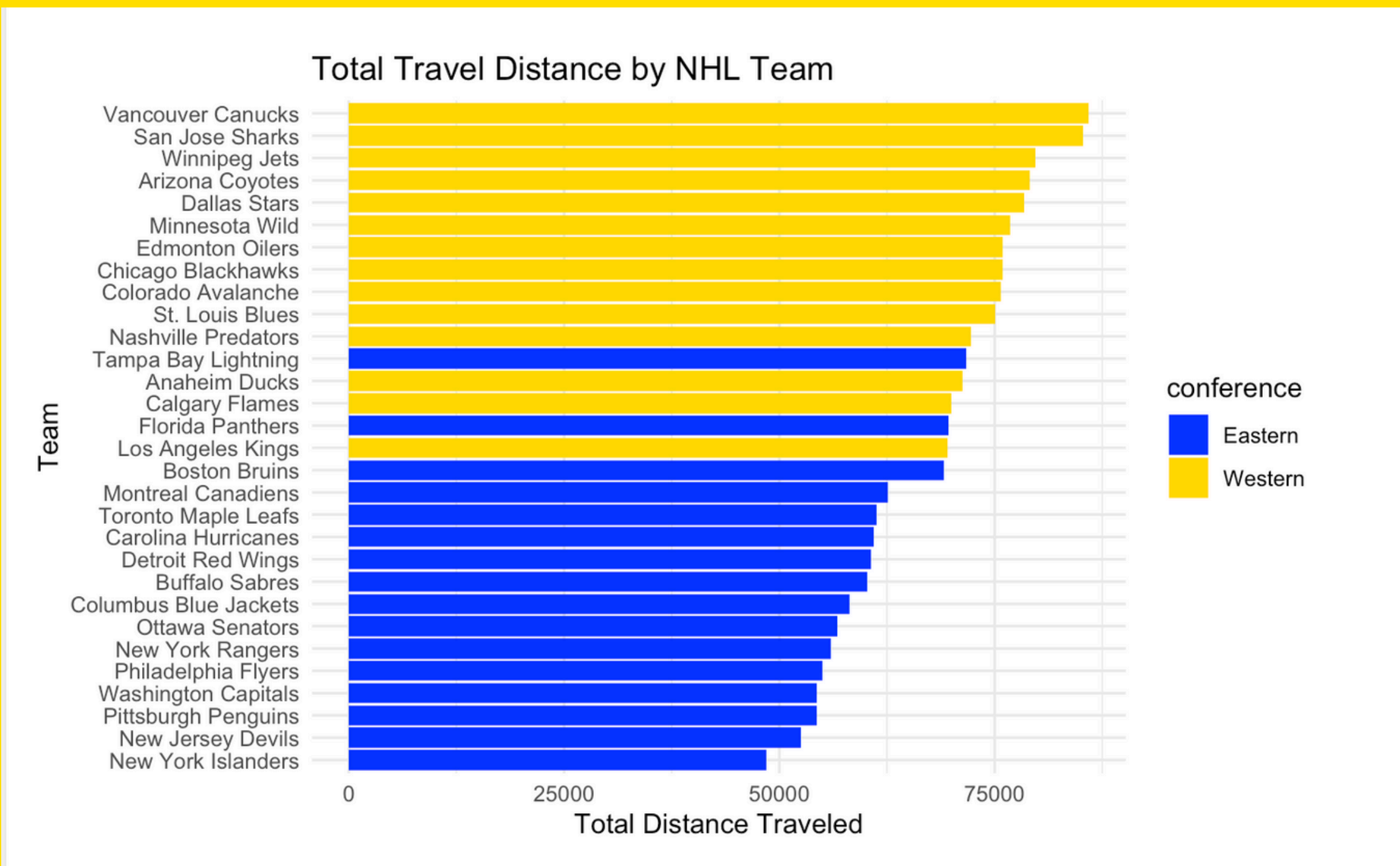
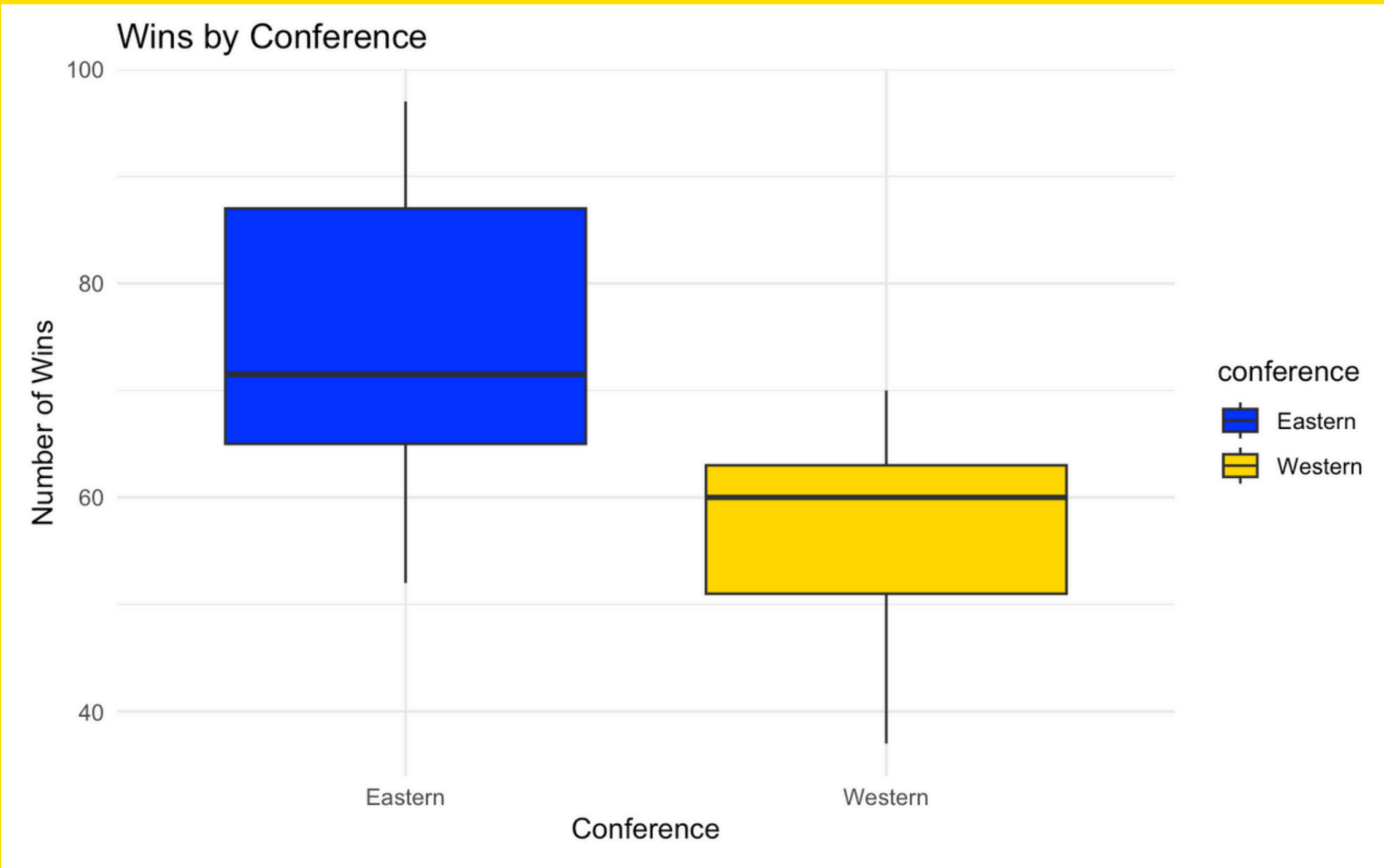
I read multiple research papers on similar projects to gain a basic understanding. I then collected data from multiple websites and entered them into Excel to clean up. After going over the data in Excel I moved the two datasets into Rstudio to feature engineer and finish cleaning up.

References

2017-18 Super Schedule - Dirk Hoag
2016-17 Super Schedule - Dirk Hoag
Game Level Data - Hockey Reference

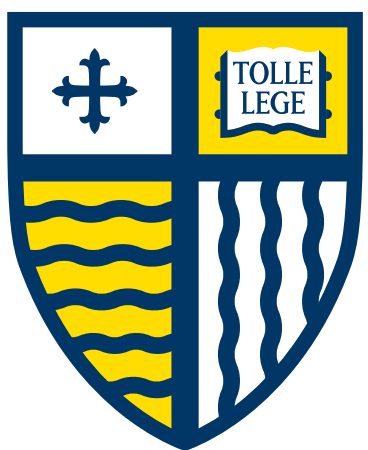
Analysis

- Distance has a statistically significant negative impact on the odds of winning. As teams travel further, their chances of winning decrease.
- Rest days and time zone difference have a slight effect on the likelihood of winning.
- The western conference travels significantly more and the eastern conference wins significantly more games.



Conclusion

While travel distance alone isn't a major factor, its interaction with time zone changes and rest appears to moderately influence team performance. Home advantage remains strong, and structural factors like conference and division assignments lead to meaningful differences in travel burdens and possibly competitive fairness.



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